
STATISTICAL PHYSICS METHODS IN COGNITIVE NEUROSCIENCE

by

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Abstract. — This is an introduction to understand brain dynamics in language of dynamical systems and statistical physics. We begin with introducing the model which is a simplistic representation of the brain in terms of the neuronal network. We then define certain order parameters or measurable quantities in terms of the avalanche or cluster size [1] and also, understand the significance of what a phase transition is. We further go on to discuss two landmark papers on the topic of brain network criticality. The first one [3] observes signatures of critical behaviour in rat cortical tissue; and the second one [4] is a hybrid study using the human connectome data, and computing the behaviour near critical point. It shows the emergence of centres of coherent brain activity responsible for major cognitive functioning at the critical point. Finally, we state some advantages and limitations of this approach to neuroscience.

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1. Introduction

In the past century, all the sciences and mathematics have made leaps of progress, but none more so than physics and its interplay with mathematics. In terms of predictive power and theoretical understanding of the fundamental laws underlying the universe, physics surpasses other sciences. The reason for the same is the existence of a theoretical strain of studies, parallel to the experimental side of progress.

In biology, such a theoretical strain is very weak, more so in neuroscience due to the immense complexity of the underlying system, the brain. For years altogether, people thought of understanding it like a black box studying the input stimuli and output response, then progress was made into the neural structures, understanding how a single neuron operates, but nothing to integrate the two vastly separated (by hierarchy) stories.

In the previous decades, the methods of non-equilibrium statistical dynamics were not well understood or developed to apply it to the brain to get meaningful information, but now, we can do so and unravel some previously unthought-of phenomenon. In this article, we shall introduce these methods and analyse the brain dynamics as a whole under the lens of statistical mechanics and phase transitions.

Key words and phrases. — Criticality, Phase Transitions, Non-equilibrium Statistical Physics, Emergent Properties, Neural Dynamics.

2. A brief History

Mathematics and Biology are two very diverging fields, but they do intermittently intersect.

D'Arcy Thomson, a fine biologist first showed his mathematical abilities in his book, '*On Growth and Form*', in which he pioneered the application of mathematics to biology. Schrodinger, the famous physicist, wrote in 1944 '*What is life?*' trying to use the mathematics developed by that time to explain biological processes. However, he concluded that biological systems are too complex to be studied mathematically. This idea has changed over the decades, as our mathematics and ability to understand statistical phenomenon developed along with our more profound understanding of biological systems. Nicolas Rashevsky was a theoretical physicist who wrote the book '*Physico-Mathematical Foundations of Biology*' and developed the first model of neural networks. Turing around the same time forwarded the idea. He, in his own terms emphasised the concept of what we now call "equipotentiality" of the cerebral cortex, and discussed the importance of development and learning in constructing a working brain. In his view, intelligence is not so much made, but grown.

Donald Hebb in his book, '*The Organization of Behaviour*' proposed a model of learning, founded on the mechanism of neural plasticity that is today known as Hebbian learning, and its later variants were early models for long term potentiation. Moreover, after a lot of experimentation and data analysis, a biophysical model of a neuron, the Hodgkin-Huxley model was proposed, where each component of an excitable cell was modelled by an electrical element. All this work around the 1950s paved the way for modern methods in neuroscience, like the one we shall be discussing in this article.

3. Hierarchical Structure and the Mapping to Mathematics

The organisation of the brain can be studied at various levels, from the patterns of firing of a single neuron to higher cognitive abilities like decision making and aspects of behaviour. In the current context we can call the activity at neuronal level, microscopic and we can choose any higher level of analysis, to be macroscopic. Mathematical models provide a mathematical analogue of each level in the hierarchy. We can use the theory of dynamical systems, non linear and stochastic (with noise). Considering this as microscopic, we can now use non-equilibrium statistical mechanics to get observables at any higher level of analysis. Statistical mechanics helps us translate from the microscopic dynamics to the macroscopic (emergent) properties.

4. Method

We model the brain with a directed $\mathbb{R}$ -coloured $\mathbb{R}$ -weighted graph

$$(V, E, h: V \rightarrow \mathbb{R}, w: E \rightarrow \mathbb{R})$$

V is the set of vertices and is supposed to represent the collection of neurons while the edges E represent the connectivity structure. The quantities $h(v), w(e)$ represent the threshold potential of the neuron v and the strength of the synapse given by the edge e , respectively. Each vertex (node) can be in one of three states $\mathcal{Q}, \mathcal{E}, \mathcal{R}$ which represents whether the neuron is *quiescent*, *excited* or *refractory*. Each node assigned a state represents a state of the system. In other words, a state of the system is a map $\sigma: V \rightarrow \{\mathcal{Q}, \mathcal{E}, \mathcal{R}\}$. Let the collection of these maps be Σ .

We now give the dynamics by the following probabilistic time evolution T defined on Σ .

$$\begin{aligned} \sigma_v = \mathcal{Q} &\rightarrow T(\sigma)_v = \mathcal{E} \text{ with probability } g(\sigma, v) \text{ -Quiescent to Excited} \\ \sigma_v = \mathcal{E} &\rightarrow T(\sigma)_v = \mathcal{R} \text{ with probability } 1 \text{ -Excited to Refractory} \\ \sigma_v = \mathcal{R} &\rightarrow T(\sigma)_v = \mathcal{Q} \text{ with probability } r_2 \text{ -Refractory to Quiescent} \end{aligned}$$

where

$$g(\sigma, v) = \begin{cases} 1 & \text{if } \sum_{e \in A(\sigma, v)} w_e > h_v \\ r_1 & \text{otherwise} \end{cases}$$

$$A(\sigma, v) = \{e \in \text{tar}^{-1}\{v\} \mid \sigma_{\text{src } e} = \mathcal{E}\}$$

Intuitively, $A(v, \sigma)$ is the set of edges into v which are coming from excited vertices $g(\sigma, v)$ is just saying that if the excitation is more than the threshold, the vertex definitely becomes excited or in the other case, it may randomly excite due to an external signal with probability r_1 . The above dynamics are equivalent to the following Markov process.

$$\Pr(X_{n+1} = \sigma_2 \wedge \sigma_2(v) = \mathcal{B} \mid X_n = \sigma_1 \wedge \sigma_1(v) = \mathcal{A}) = \left(\begin{array}{c|ccc} \mathcal{A} \downarrow, \mathcal{B} \rightarrow & \mathcal{Q} & \mathcal{E} & \mathcal{R} \\ \hline \mathcal{Q} & 1 - g(\sigma_1, v) & g(\sigma_1, v) & 0 \\ \mathcal{E} & 0 & 0 & 1 \\ \mathcal{R} & r_2 & 0 & 1 - r_2 \end{array} \right) = p\left(\mathcal{A} \xrightarrow{v} \mathcal{B} \mid \sigma_1\right)$$

Thus we have the following transition matrix

$$\begin{aligned} \Pr(X_{n+1} = \sigma_2 \mid X_n = \sigma_1) &= \prod_{v \in V} p\left(\sigma_1(v) \xrightarrow{v} \sigma_2(v) \mid \sigma_1\right) \\ &= P(\sigma_1 \rightarrow \sigma_2) \end{aligned}$$

We define the following quantities

- Activity of a state σ : $a(\sigma) = \text{card } \sigma^{-1}\{\mathcal{E}\}$. That is, it is the number of excited neurons in a state.
- An avalanche with minimal activity $a_{\min}$, maximal activity (or size) $a_{\max}$ and duration t is defined at be an event

$$X_n = \sigma_0, X_{n+1} = \sigma_1, \dots, X_{n+t} = \sigma_t, X_{n+t+1} = \sigma_{t+1}$$

such that

- $a(\sigma_0), a(\sigma_{n+t+1}) < a_{\min}$
- $a(\sigma_{n+i}) \geq a_{\min}$ for $1 \leq i \leq t$
- $\max_{1 \leq i \leq t} a(\sigma_{n+i}) = a_{\max}$

Avalanches are an object of interest, correspond to bursts of activity in the brain.

We define the following variables

$$N_{\text{size}}(a_{\min}, a_{\max} \mid X_0 = \sigma, n) = \langle \text{Number of avalanches of minimal activity } a_{\min} \text{ and } \mathbf{size} \ a_{\max} \text{ in the sequence } X_0 = \sigma, \dots, X_n \rangle$$

$$N_{\text{time}}(a_{\min}, t \mid X_0 = \sigma, n) = \langle \text{Number of avalanches of minimal activity } a_0, \mathbf{duration} \ t \text{ in the sequence } X_0 = \sigma, \dots, X_n \rangle$$

$$\Gamma(a_{\min}, t \mid X_0 = \sigma, n) = \langle \mathbf{Average size} \text{ of avalanches of minimal activity } a_{\min} \mathbf{duration} \ t \text{ in the sequence } X_0 = \sigma, \dots, X_n \rangle$$

We define the thermodynamic quantities

$$\begin{aligned} f_{\text{size}}(a_{\min}, a_{\max}) &= \left\langle \lim_{n \rightarrow \infty} \frac{N_{\text{size}}(a_{\min}, a_{\max} \mid X_0 = \sigma, n)}{n} \right\rangle_{\sigma} \\ f_{\text{time}}(a_{\min}, t) &= \left\langle \lim_{n \rightarrow \infty} \frac{N_{\text{time}}(a_{\min}, t \mid X_0 = \sigma, n)}{n} \right\rangle_{\sigma} \\ \gamma(a_{\min}, t) &= \left\langle \lim_{n \rightarrow \infty} \Gamma(a_{\min}, t \mid X_0 = \sigma, n) \right\rangle_{\sigma} \end{aligned}$$

The physical meanings of these quantities are:

- $f_{\text{size}}(a_0, s)$ is the probability an avalanche of minimal activity a_0 and size s occurs
- $f_{\text{time}}(a_0, m)$ is the probability an avalanche of minimal activity a_0 and time m occurs
- $\gamma(a_0, m)$ is the average size of an avalanche of minimal activity a_0 that lasts for a time m

When a system goes through a phase transition, the points where thermodynamic quantities show non-analytic behaviour. These points, called critical points are characterised by:

- *Power Law Behaviour*: All thermodynamic quantities have non-analytic behaviour given by power laws.
- *Scaling Behaviour* The exponents of various thermodynamic quantities are related to each other by simple relations.

- *Scale Invariance* Zooming in or out in the scales retain the behaviour, there is a self-similarity is a characteristic.
- *Universality* Many vastly different systems have the same behaviour at the critical point.

5. Literature

5.1. Criticality in Rat Cortical Tissue [3]. — The tasks a neuronal network performs are highly complex and diverse and it is hypothesised that to function optimally, the network operated at a critical point. In this study [3] study the dynamics of cultured cortical networks of rat brains and analyse the data collected at the level of single neurons using the framework of nonequilibrium phase transitions.

They removed 400 μm thick slices of cortical tissue from living rats, cultured them until they reached maturity (approximately one month), and placed them on an array of 512 electrodes spaced 60 μm .

Fig 3 shows samples change from having early exponential cutoffs, to straightforward power laws, to having humps in their distribution. The graphs show that different samples exist at different points in the phase diagram. Factors that determine the phase of a given sample include number of connections (or the edges E in our graph, and connection strength w . Computations can move along the phase diagram by increasing connection strength. The experiment used drugs to alter network connectivity and hence, to move the network to subcritical and supercritical.

Increasing the number or the strength of connections in a network moves the network toward supercriticality. For mature cortical tissue, departures from criticality may correspond to pathological states. Recent studies suggest that epilepsy is one such state.

The power-laws go as: $f_{\text{size}}(0, a_{\text{max}}) \sim a_{\text{max}}^{-\alpha}$, $f_{\text{time}}(0, t) \sim t^{-\tau}$ and $\gamma(0, t) \sim t^{-\frac{1}{\nu}}$

The relationship between the ‘critical exponents’:

$$\frac{\alpha - 1}{\tau - 1} = \frac{1}{\nu}$$

This relationship is theoretical and is consistent with observed values of the exponents.

An analysis of criticality in cortical tissue is significant because of its relation to optimal information processing, information storage, dynamic response, and computation [5]. The collapse of avalanche data onto a universal scaling function as predicted by the theory of dynamic critical phenomena provides a clear demonstration of quantitative universality in our system which is a key pointer to criticality.

It is concluded that critical dynamics depend intricately on a unique network structure.

The study highlights the fact that we can develop a diagnostics for identifying the critical network from non-critical. Studies are also being done on how criticality emerges on self-organization of neuronal networks and how self-organization is always directed towards optimal processing power. Observing the network in various stages of maturation can also help us understand the universal features of critical networks.

5.2. Resting State Networks emerge at Criticality on a model of Human Connectome [4].

— In this study they approach the problem of finding a relation between large-scale brain structure and function, by studying the dynamical regime, under which realistic spatio-temporal patterns of brain activity emerge in the computation, from the empirically derived network of human brain neuro-anatomical connections. The results highlight *critical dynamics unfolding* on the structural connectivity of the human brain.

Here, for our underlying graph/ network we import the data from a human connectome. Data is obtained from the white matter tracts connecting a parcellation of 998 nodes covering the entire human brain, and the weights of the strength depend on how well connected the two nodes are.

As depicted in Fig 7, the dynamics of the model show a transition as a function of the threshold h , which which is a parameter we have attached to each node. For relatively small values of h even the weakest connections are enough for the activity to spread, resulting in a regime with a relatively high activity level. On the contrary, for high values of h the activity only flows through the few strongest connections and therefore the overall activity decreases. At an intermediate value, a critical point h_c can be identified by the peak in the size of the second largest cluster. This also what is generally done in fMRI studies.

The idea is to understand spontaneous brain dynamics at the large scale organize into relatively few important spatiotemporal patterns. This is important because a wide range of experiments fMRI have emphasized that these spatial clusters of coherent activity, termed **Resting State Networks**

(RSN)[2] are specifically associated with neuronal systems responsible for sensory, cognitive, and behavioral functions. The pattern of correlations in these networks has been shown to change with various cognitive and pathophysiological conditions.

The RSN activity exhibits peculiar scaling properties, resembling dynamics near the critical point. This study is a landmark because this is a hybrid modeling approach (realistic anatomical connectivity plus a simple dynamical rule). This approach captures relevant spatiotemporal aspects of brain dynamics, matching with the fMRI observations if the dynamical regime is critical.

6. Advantages and Limitations

There are two ways of understanding cognitive systems mathematically. The first one is meant to demonstrate that we understand some particular aspects of the system, for example, the Hodgkin-Huxley model of the neuron; and the other is a broader approach it tries to discover new structural and functional relationships in the system.

The former allows numerical simulation and verification, while the latter attempts to capture the organising principles underlying the behaviour of the system. While the former is data-driven, the latter is driven by the creative act of the modeller and ultimately endorsed by experimentation.

Both of these methods come into the realm of mathematical methods.

We can put to our use many methods already developed in physics such as non-equilibrium statistical physics, quantum field theory, Renormalisation Group theory and others. to understand the dynamical properties of the brain as a whole.

These methods have equipped us with ways to compute n -point functions which can give a theoretical backing to correlators observed from fMRI studies and further enhance our structure-to-function understanding of the brain. These mathematical ideas give solid proofs for the optimal and efficient functioning of the brain, which earlier could only be approached by evolutionary reasoning. We can hope to understand the behaviour as an emergent property of the brain as a whole.

At the same time, the models we are currently choosing are simplistic and hence cannot give a complete understanding of brain dynamics as incorporating all aspects will make the model too complicated to work with analytically (with present methods). Another issue is the communication gap between streams of analytical subjects like mathematics, physics, theoretical computer science and soft subjects like neuroscience and psychology; which slows down the absorption of methods and understanding each other's fields.

References

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Appendix A

Notation

- $\wedge$ - AND
- $\Pr(A|B)$ - Probability of occurrence of Event A given the occurrence of Event B
- X_n - Random Variable denoting the state of the system (brain) at time n
- card - Cardinality or Size of the set

Appendix B

Figures

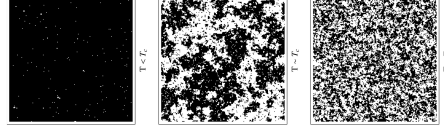


FIGURE 1. Representation of Subcritical, Critical and Super-Critical Activity

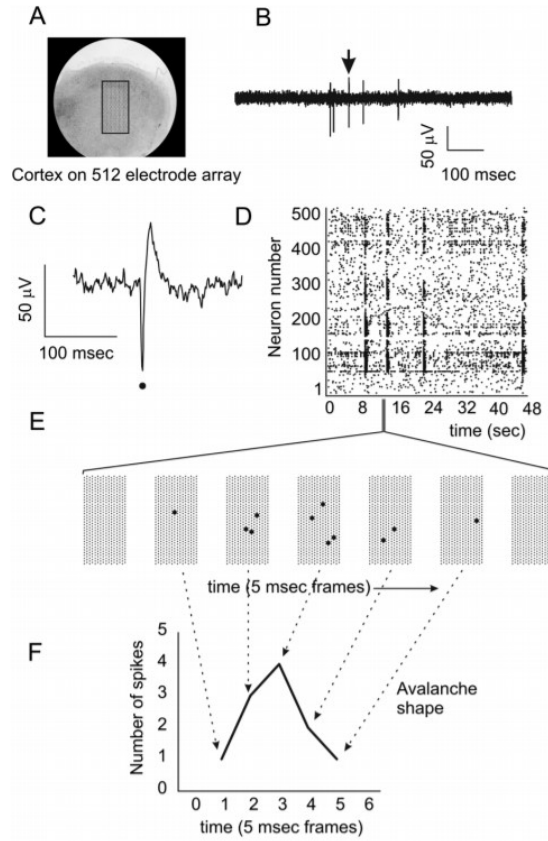


FIGURE 2. **A.** Micrograph of cultured cortical slice on a 512 electrode array; **B.** Voltage trace from one electrode; **C.** Time of spike is marked by black dot; **D.** Raster plot of spike times (dots) from many neurons over a 48 s interval; **E.** Expanded view of network activity reveals an avalanche. Each frame represents the array at one 5 ms bin. Small dots are electrode locations; large dots are spikes on array **F.** Avalanche shape is obtained by plotting the number of spikes in each frame versus time

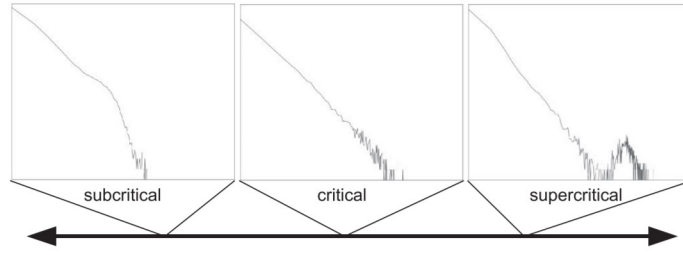


FIGURE 3. Avalanche size histograms from three different samples of rat cortex. The horizontal line represents a path in some representative (such as increasing mean connection strength or increasing number of connections) direction in the complex parameter space of network, representing how critical the sample of interest is.

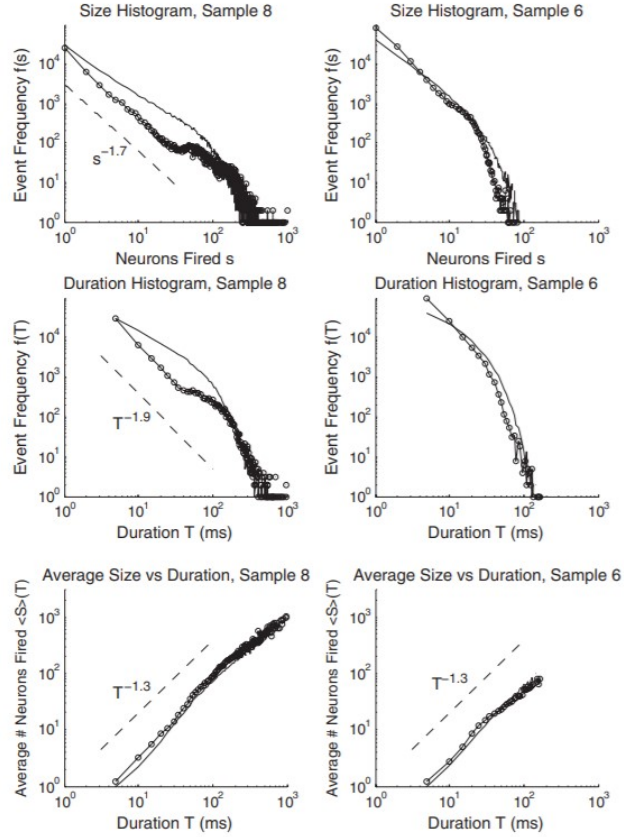


FIGURE 4. Contrasting size and duration distributions and average size for fixed duration from a critical data set (left) and a subcritical data set (right). The dashed lines on the left (near critical) column correspond to power laws with exponent 1.7, 1.9, and 1.3, corresponding to the critical exponents α , τ , and $\frac{1}{\nu}$ respectively.

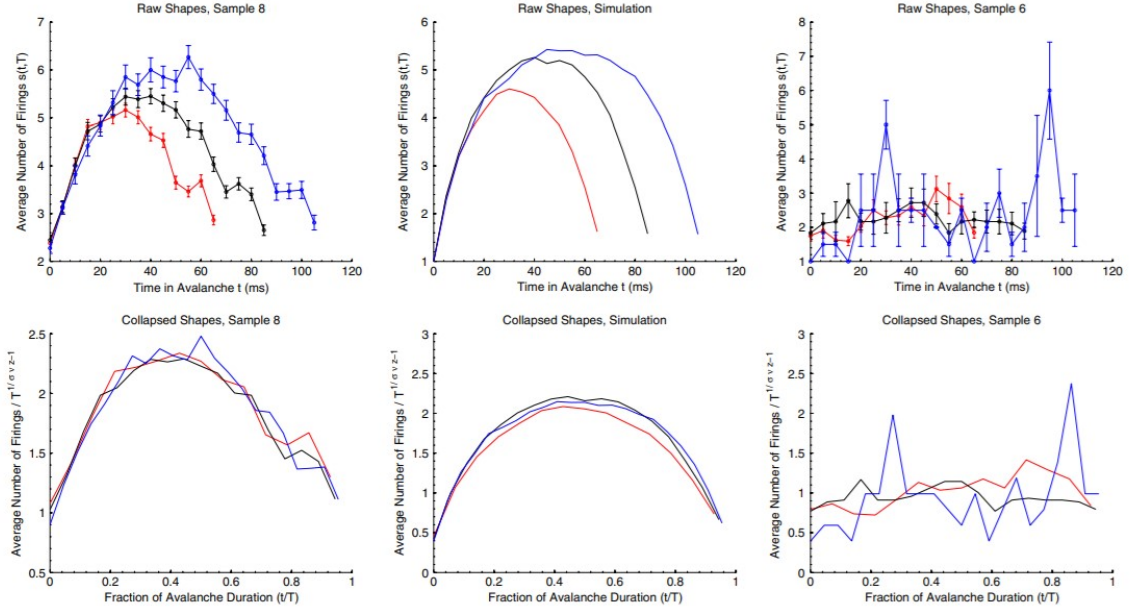


FIGURE 5. Avalanche shape collapses. Shapes and attempted collapses from three data sets: two experimental and one simulated. Shapes are produced by averaging the temporal profiles of all avalanches of a particular duration; different colors here represent different durations. The collapses are plotted by rescaling the horizontal and vertical axes. The left- and rightmost data correspond to experimental data close to and far from criticality, respectively

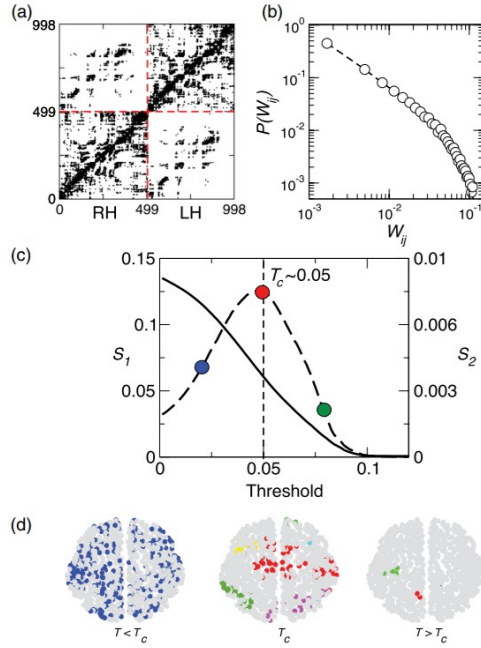


FIGURE 6. The model. (a) shows the model adjacency matrix (i.e., the connectome), (b) shows the edge weights distribution. (c) shows the giant cluster's size (i.e., the order parameter, S_1 , solid line) and the second largest cluster's size (S_2 , dashed line) as a function of the threshold h (control parameter) as well as the critical point $h_c = 0.05$. (d) illustrates examples of clusters (denoted with different colors) at the three (sub-critical, critical and super-critical) h values denoted with the filled circles in panel (c)

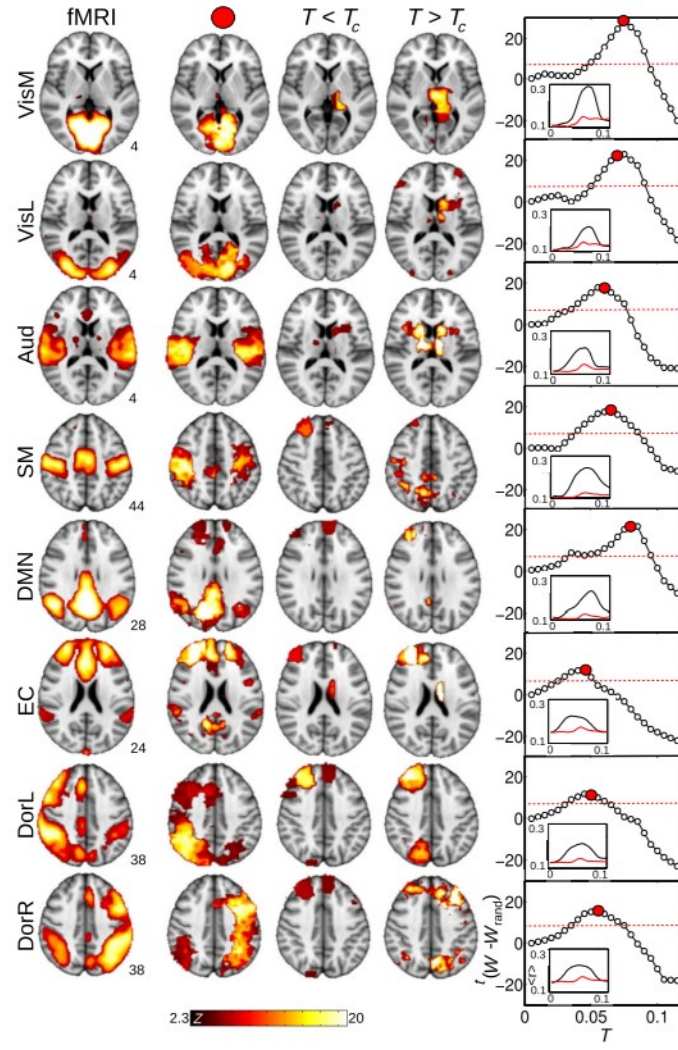


FIGURE 7. The spatial organization of the human brain RSN emerges spontaneously in the model near h_c . RSN obtained from the model are shown for three values of $h = h_c, h < h_c(0.03), h > h_c(0.1)$, and the correlation maxima with the human RSN fMRI, which matches with h_c